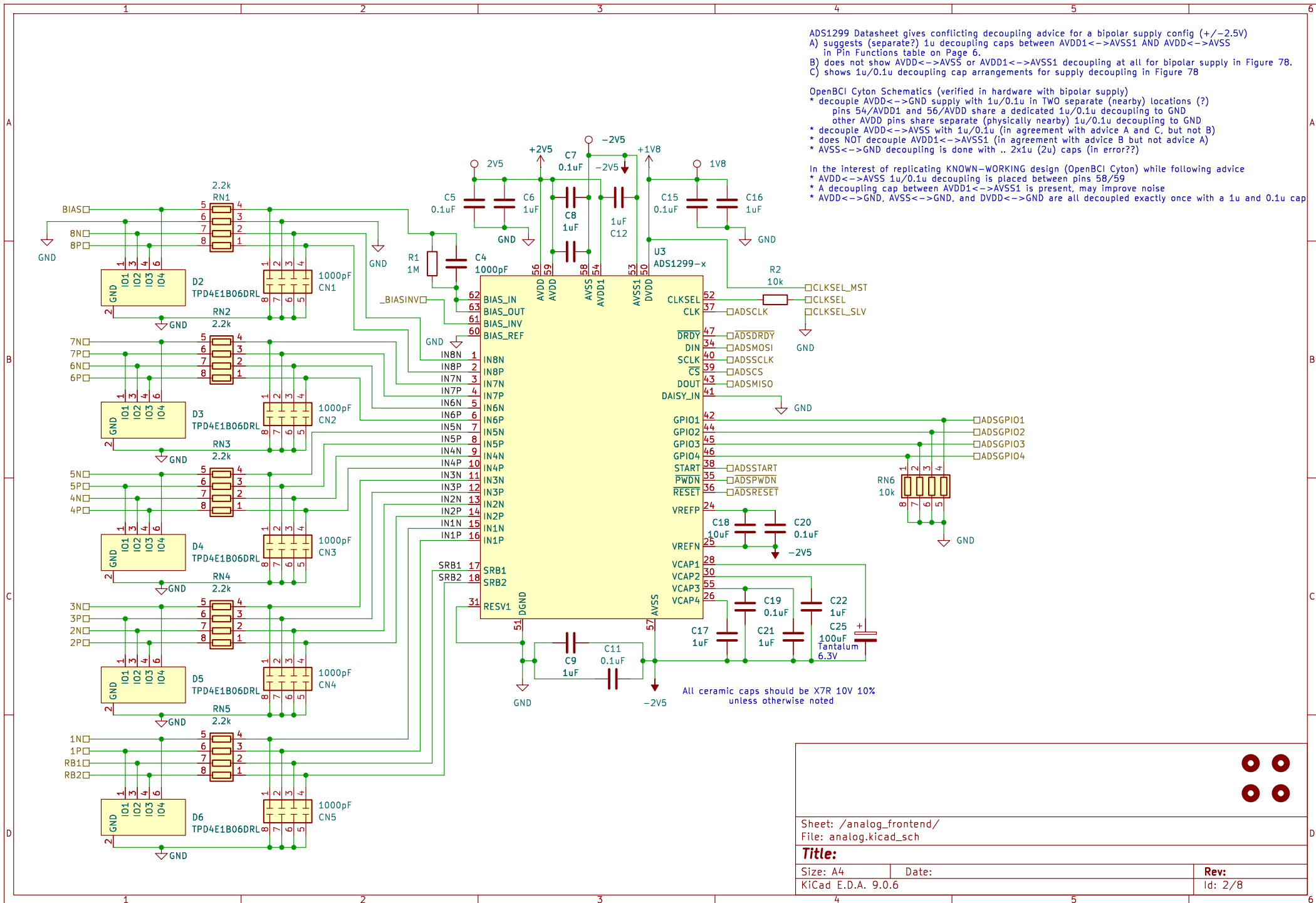




Sheet: /analog_frontend/
File: analog.kicad_sch

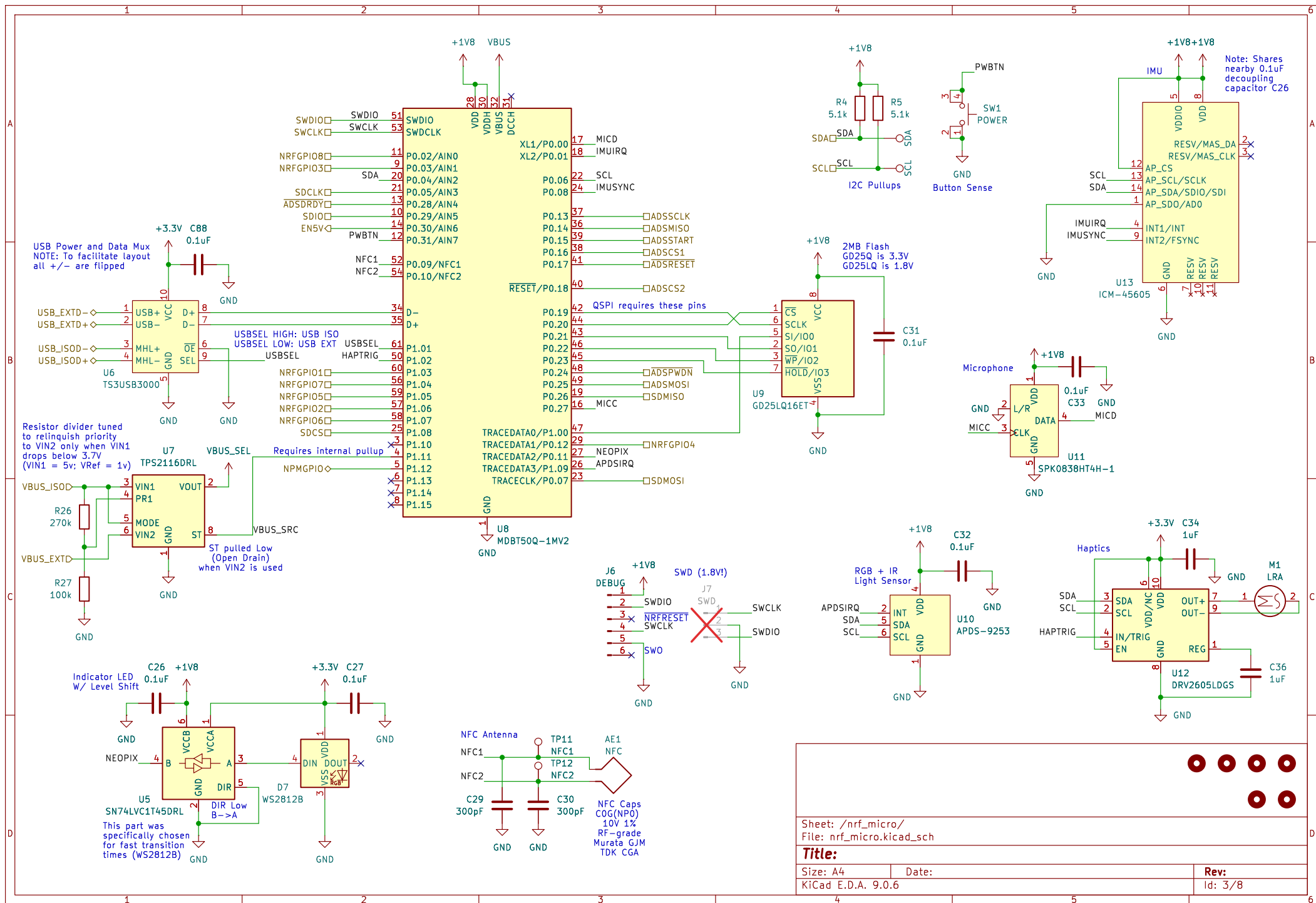
Title:

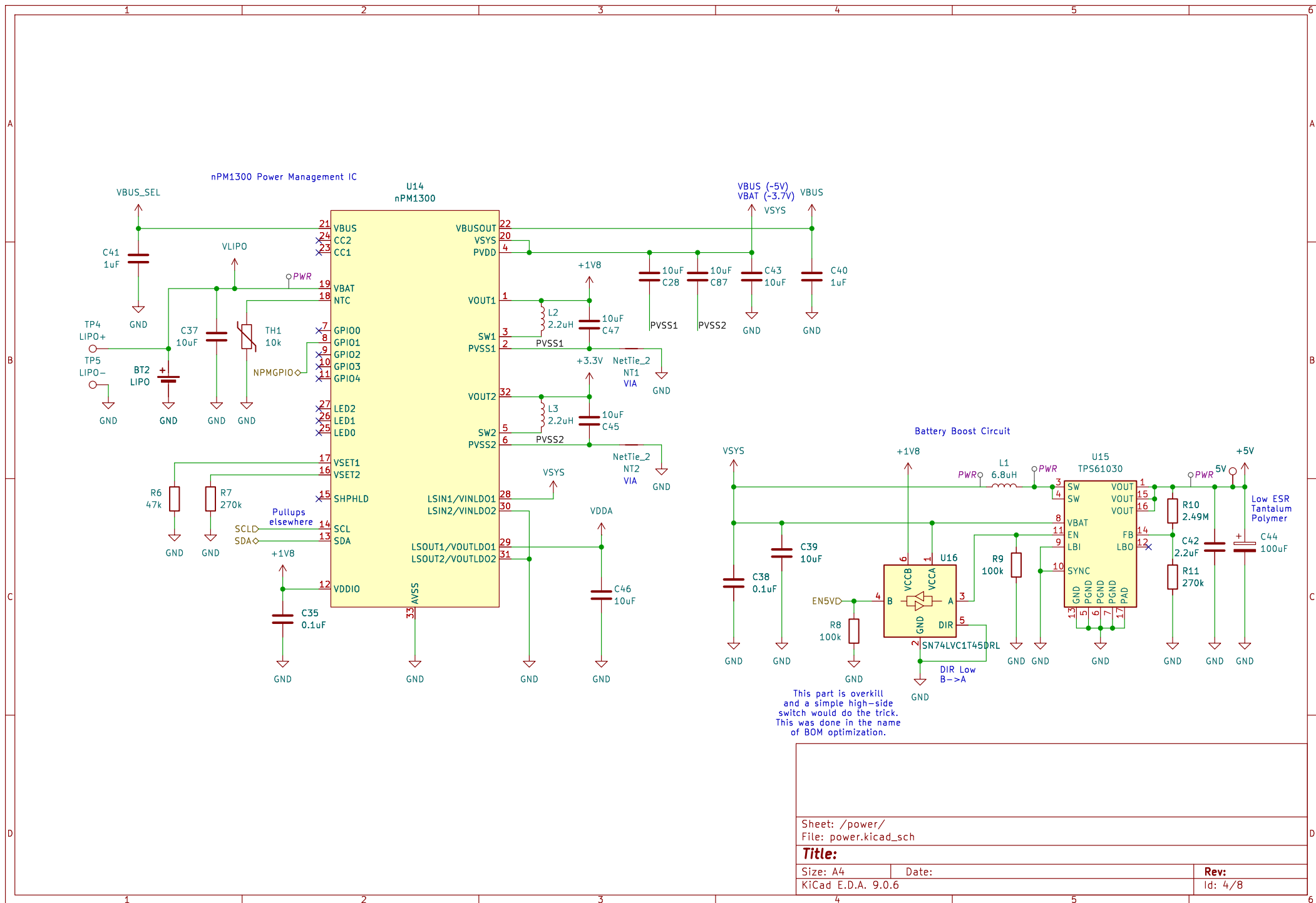
Size: A4 Date:

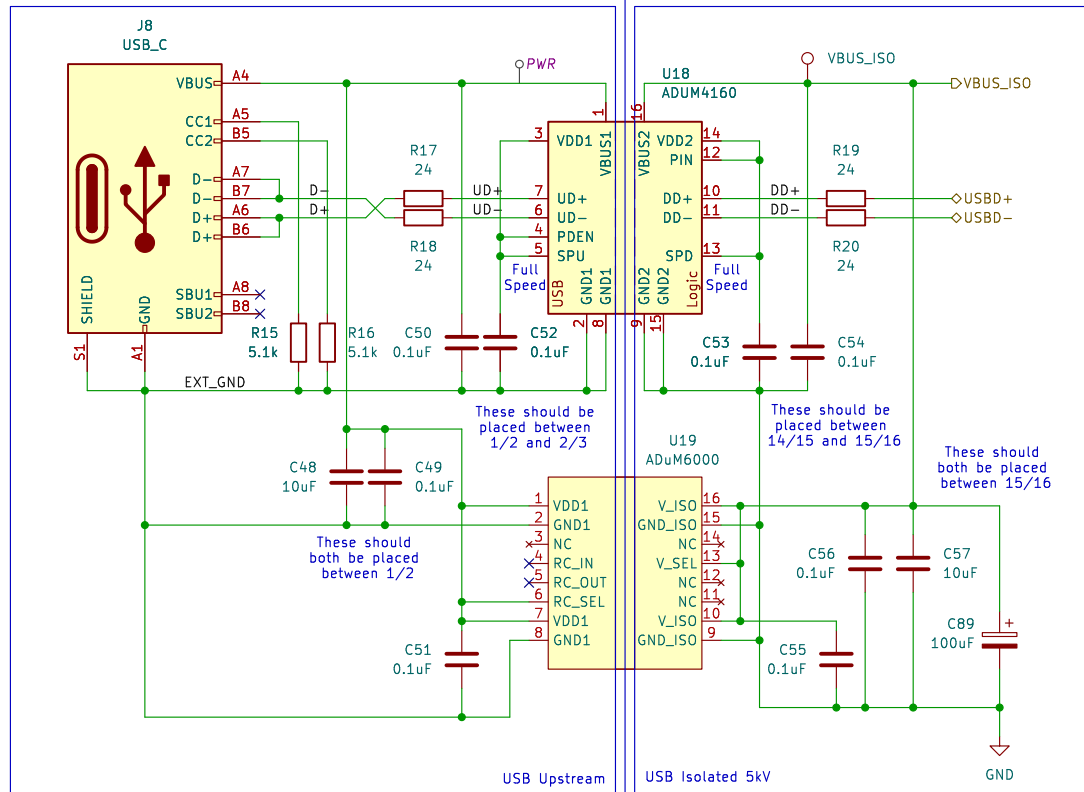
KiCad E.D.A. 9.0.6

Rev:

Id: 2/8







ADuM6000 provides MAXIMUM 80 mA of current.
It gets VERY hot when doing so.

Recommend keeping I_{load} from VBUS at maximum 50mA.
External hardware connected to supplied power rails should
be tuned to keep current draw limited, or else ADuM6000
could get hot enough to damage battery if in proximity

Schematic based on Adafruit USB Isolator
<https://github.com/adafruit/Adafruit-USB-Isolator-PCB>
Substituted ADUM6000 in place of ADUM5000
ADUM6000 provides 80mA/5kV isolation instead of 100mA/2.5kV
Limited to USB Full-Speed (12 Mbps)
Limited to ~80ma supply current on VBUS

Sheet: /usb_isolation/
File: usb_isolation.kicad_sch

Title:

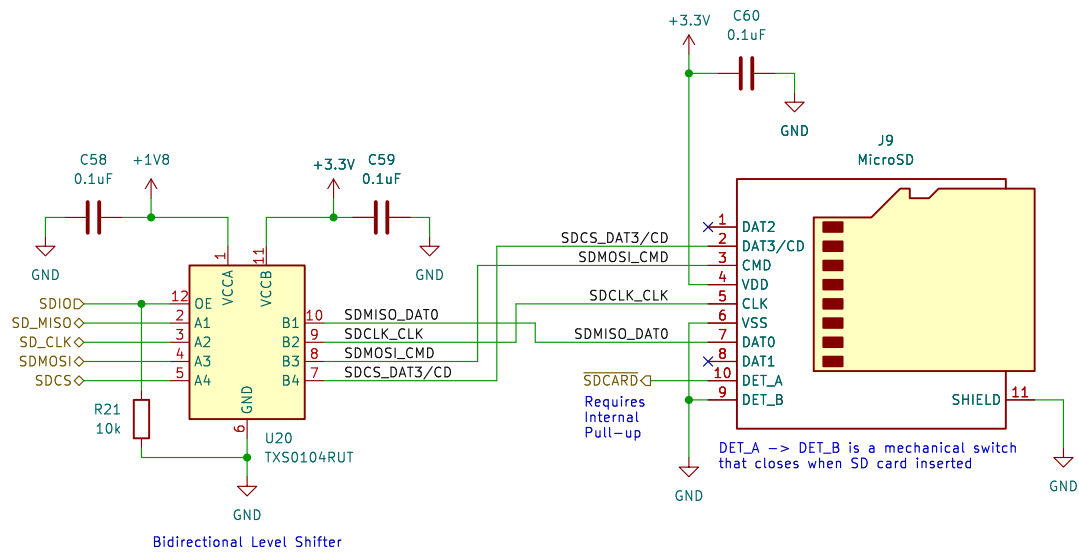
Size: A4

Date:

KiCad E.D.A. 9.0.6

Rev:

Id: 5/8



Sheet: /sd_card/
File: sd_card.kicad_sch

Title:

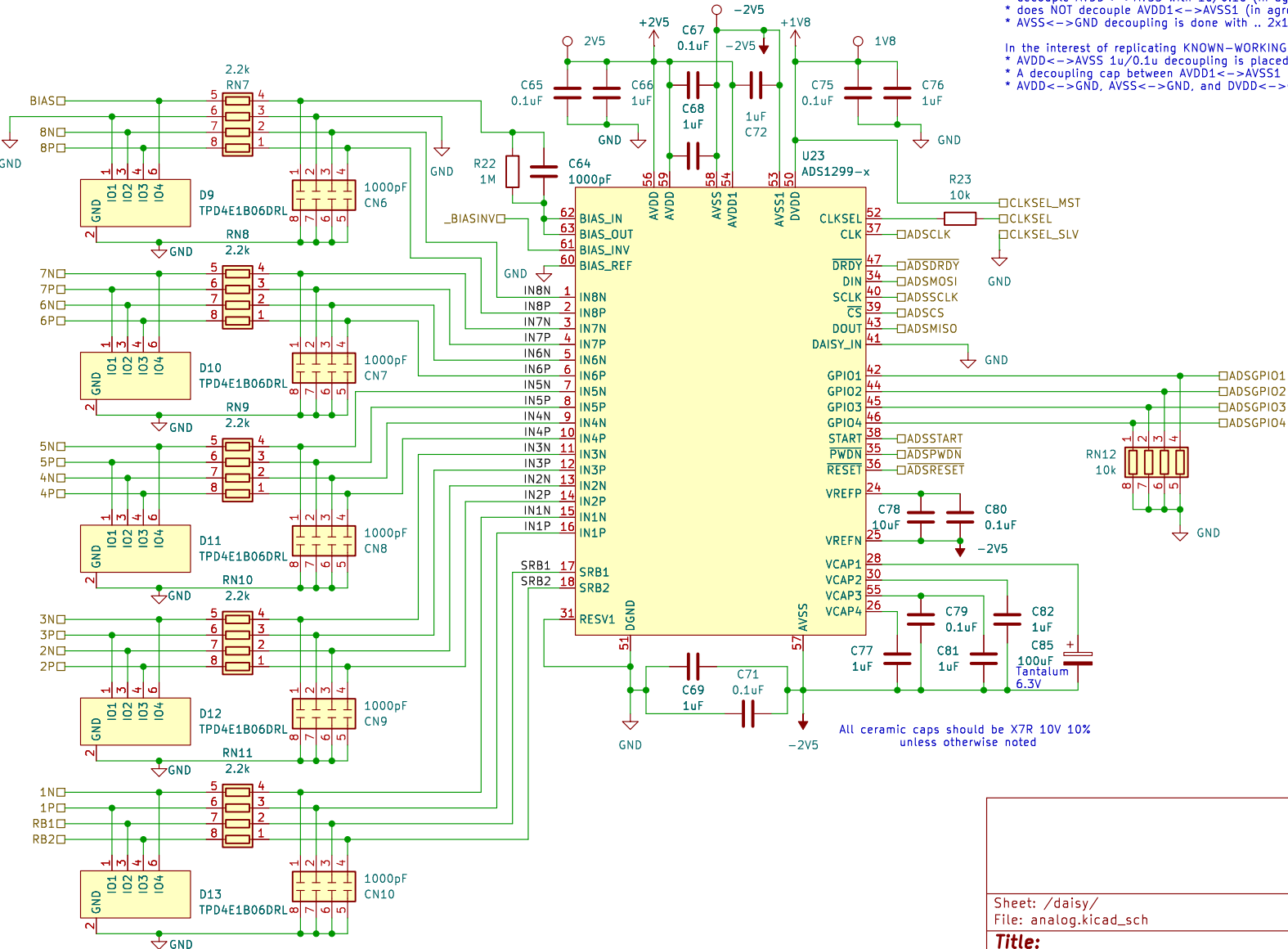
Size: A4

Date:

KiCad E.D.A. 9.0.6

Rev:

Id: 6/8



ADS1299 Datasheet gives conflicting decoupling advice for a bipolar supply config (+/-2.5V)
 A) suggests (separate?) 1u decoupling caps between AVDD1<->AVSS1 AND AVDD<->AVSS
 in Pin Functions table on Page 6.
 B) does not show AVDD<->AVSS or AVDD1<->AVSS1 decoupling at all for bipolar supply in Figure 78.
 C) shows 1u/0.1u decoupling cap arrangements for supply decoupling in Figure 78

OpenBCI Cyton Schematics (verified in hardware with bipolar supply)
 * decouple AVDD<->GND supply with 1u/0.1u in TWO separate (nearby) locations (?)
 pins 54/AVDD1 and 56/AVDD share a dedicated 1u/0.1u decoupling to GND
 other AVDD pins share separate (physically nearby) 1u/0.1u decoupling to GND
 * decouple AVDD<->AVSS with 1u/0.1u (in agreement with advice A and C, but not B)
 * does NOT decouple AVDD1<->AVSS1 (in agreement with advice B but not advice A)
 * AVSS<->GND decoupling is done with .. 2x1u (2u) caps (in error??)

In the interest of replicating KNOWN-WORKING design (OpenBCI Cyton) while following advice
 * AVDD<->AVSS 1u/0.1u decoupling is placed between pins 58/59
 * A decoupling cap between AVDD1<->AVSS1 is present, may improve noise
 * AVDD<->GND, AVSS<->GND, and DVDD<->GND are all decoupled exactly once with a 1u and 0.1u cap

All ceramic caps should be X7R 10V 10%
 unless otherwise noted

Sheet: /daisy/
 File: analog.kicad_sch

Title:

Size: A4

Date:

KiCad E.D.A. 9.0.6

Rev:

Id: 7/8

